

SANDIP SHAW

AI Developer

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Professional Summary

AI Developer with 2 years of experience across AI, Data Science and analysis, building python based agentic workflows and Generative AI solutions that automate business processes. Hands-on with Crew AI and LangChain based Agentic RAG systems including prompt engineering, API development and integration, and full AI/ML lifecycle that includes model development and training and containerised deployment on AWS. Comfortable in working with Git-based version control with CI/CD pipelines (GitHub Actions), applying modular design, code readability and unit testing to delivered code. Proficient in SQL and Python.

SKILLS

GenAI & Agentic AI: Generative AI, Prompt Engineering, LangChain, Agentic Workflows, CrewAI, RAG, Hugging Face, NLP

Machine Learning: Scikit-Learn, PyTorch, TensorFlow, Keras, Deep Learning, AI/ML Lifecycle, Model Interpretability and Evaluation.

Programming and Database: Python, SQL, MySQL, Postgres, MongoDB

Engineering and Deployment: API Development & Integration, Docker, AWS, Git, GitHub, CI/CD (GitHub Actions), Unit Testing, Modular Design.

Soft Skill: Stakeholder communication, Problem structuring, Cross-functional collaboration, Ownership & delivery

EXPERIENCE

PCC Softech India | Pune, India

1. AI Developer | July 2026 – Present

- Engineered a multi-agent data ingestion system in Python using LangChain, orchestrating three agents across parsing, validation and schema normalization to process PDFs, CSVs, scanned images from nine business domains into PostgreSQL database reducing manual effort by 24+ hours per week.
- Built an IT asset benchmarking agent using CrewAI coordinating specialized agents that call web search and a vendor pricing API to derive benchmark prices for 3,000+ IT assets against service life thresholds and performed manual validation step on agent output prior to delivery.
- Containerised Python services with Docker and shipped them to AWS through the team's GitHub Actions CI/CD pipeline, operating in Git branch and pull requests workflow with peer code review, structured code into modular, unit-tested components and resolved build and deployment failures.

2. Data Scientist AI Developer Intern | January 2026 – June 2026

- Improved factual consistency of Generative AI recommendation system to 88% through prompt engineering and output anchoring in Python, validated by repeated generation testing that compared extracted device configurations across runs independent of phrasing.
- Developed 25+ REST API endpoints in Python using FastAPI and Pydantic across two billing services, designing the SQL schema across 20+ data models in PostgreSQL with SQLAlchemy and Alembic-managed migrations.
- Implemented TOTP-based two-factor authentication with Fernet envelope encryption, versioned key rotation and single-use backup codes to gate privileged billing operations, integrated with Avalara AvaTax API for tax calculation, and structured the platform into modular services covered by 350+ unit tests in pytest.

Data Analyst | Innover Digital | June 2022 – April 2023 | Kolkata, India

- Computed route-level quote pricing from 100,000+ trip records in MySQL using multi-table joins, CTEs and window functions to factor goods type and distance into per-route cost, publishing refreshed prices to the production database consumed directly by the quoting team, preventing quotes being issued against outdated costs.
- Built and maintained automated ETL and reporting pipelines serving 20–100 production users, taking data products from design through demo cycles to deployment in a Dockerised environment.
- Owned data validation and quality assurance for automated parsing of user-uploaded CSVs across 50+ tables, profiling data in Python (pandas) and enforcing quality checks before downstream reporting and analysis.

Data Analyst Intern | Krop Gains | August 2021 – March 2022 | Remote

- Built an object detection model using YOLO in PyTorch to locate diseased regions in crop imagery, training on 6,000 augmented images across 12 crop categories and implementing data and model validation pipelines, achieving 79% mAP on held-out test data.
- Developed an XGBoost regression model in Python to forecast multi-year price trends across 13 agricultural commodities, engineering features from 100GB of scraped and cleaned structured and unstructured agricultural data, achieving an R^2 of 0.87 on held-out data.

PROJECTS

AI-Powered Safe Route Recommender [Deployed]

- Developed a crime-aware route recommender for commuters using FastAPI, a KDE risk surface, and a LightGBM ensemble (70/30 blend, PR-AUC-0.92).
- Built and deployed a crime-aware route recommender returning safety-optimised routes between user-specified origin and destination, serving a LightGBM ensemble over a KDE risk surface through a FastAPI backend containerised on Azure Container Apps with a Vercel-hosted React frontend and automated CI/CD.
- Added a retrieval-augmented generation (RAG) assistant over 120,000 structured incident records using LangChain, BGE-small sentence-transformer embeddings and a FAISS vector store with overlapping chunking, applying prompt engineering to ground responses in retrieved records.

EDUCATION

MSc Data Science & Spatial Analytics - Symbiosis Institute of Geoinformatics - 06/2024 – 08/2026 | Pune | **CGPA 8.99**

BSc Data Science - Techno India University - 06/2019 – 08/2022 | Kolkata | **CGPA 8.41**

CERTIFICATIONS

- PW-Skills Full Stack Data Science Pro ([LINK](#))